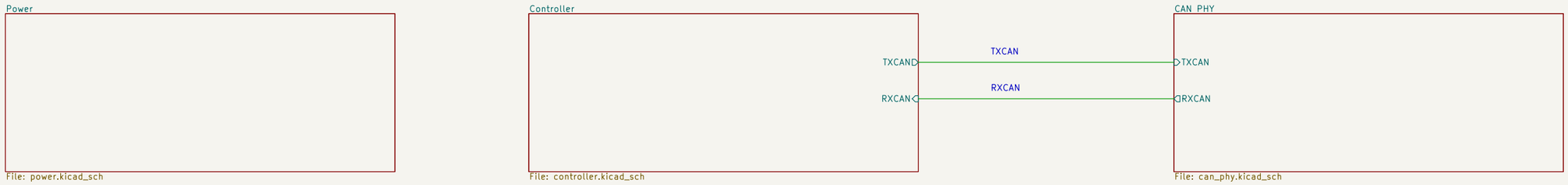
MCP2515 CAN Bus Module — Rev A — 2026-11-02

Block map: (1)(2) Power -> power.kicad_sch | (3)(4)(7) Controller -> controller.kicad_sch | (5)(6) CAN PHY -> can_phy.kicad_sch



Design notes

1. SJ1 (VIO <- +3V3): close ONLY when no host drives J2-VIO.

2. JP1: fit BOTH shunts (1-2, 3-4) to enable 120 ohm split termination; remove both to disable.

3. Input: 7-16 V DC, centre-positive. Reverse polarity blocked by D1; sustained >18 V trips F1.

4. MCP2515 16 MHz bit timing (CNF1/CNF2/CNF3): 1M 00/83/01 | 500k 00/AE/01 | 250k 01/AE/01 | 125k 03/AE/01. Disable CLKOUT (CANCTRL.CLKEN=0).

5. L2 DNP: R16/R17 0 ohm fitted. If L2 fitted, remove R16/R17.

6. R4 (OSC2 series) 0 ohm; fit ~1 kohm only if crystal overdrive measured at VIO 5 V.

Mechanical



Sheet: Root — sheet index, design notes, mechanical
9 V in · host-referenced VIO 3.0-5.5 V · JLCPCB

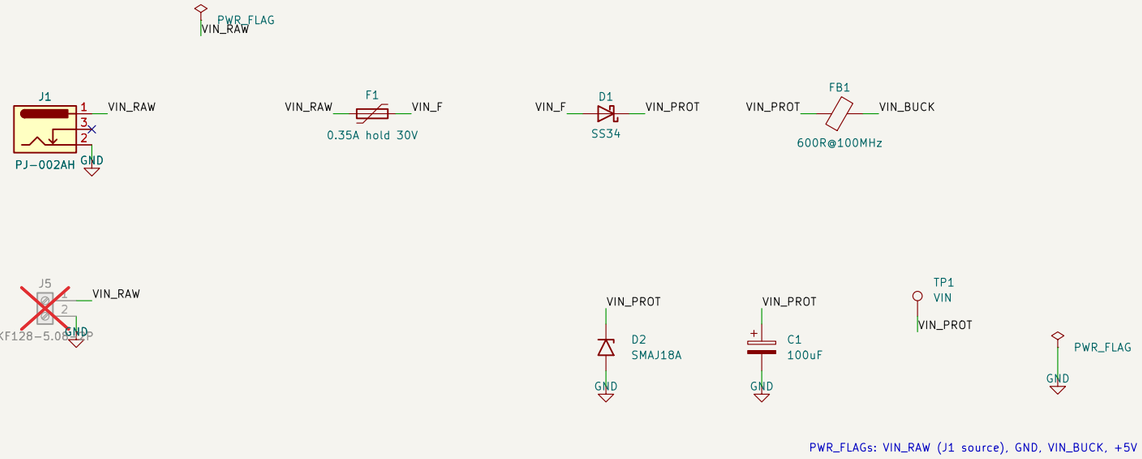
Sheet: /
File: MCP2515_CAN.kicad_sch

Title: MCP2515 CAN Bus Module

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/4

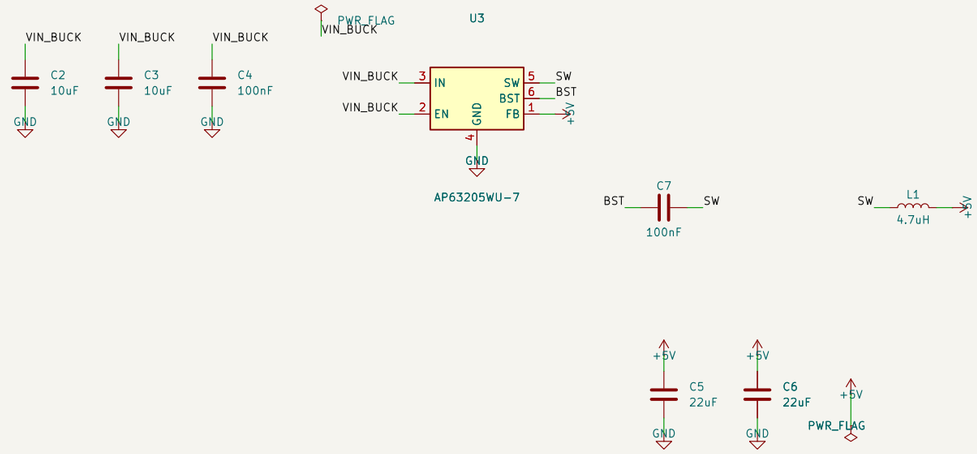
(1) Power input & protection

Input 7–16 V DC, centre-positive. Chain: J1/J1B -> F1 PTC -> D1 reverse-polarity -> (D2 TVS, C1 bulk) -> FB1 -> buck input



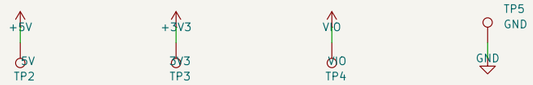
PWR_FLAGs: VIN_RAW (J1 source), GND, VIN_BUCK, +5V

(2) Voltage regulation: 5 V buck

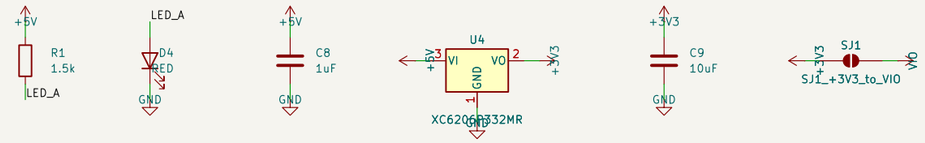


FB (U3 pin 1) tied to +5V at C5/C6 (Kelvin). EN tied to VIN_BUCK.

Rails & test points



(2) Regulation: +5V indicator, optional +3V3 LDO, VIO option



SJ1 (VIO <- +3V3): close ONLY when no host drives J2-VIO.

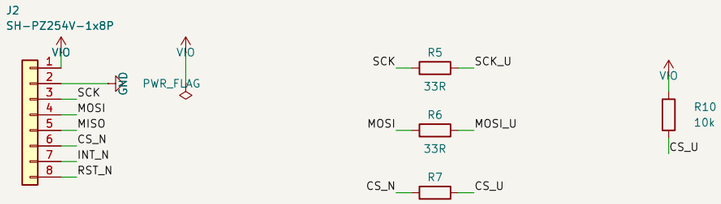
Sheet: 1 Power input & protection, 2 Voltage regulation
9 V in · host-referenced VIO 3.0-5.5 V · JLCPCB

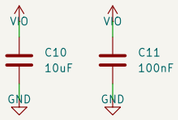
Sheet: /Power/
File: power.kicad_sch

Title: MCP2515 CAN Bus Module

Size: A3	Date: 2026-11-02	**Rev: A**
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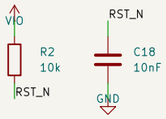
(7) SPI host interface







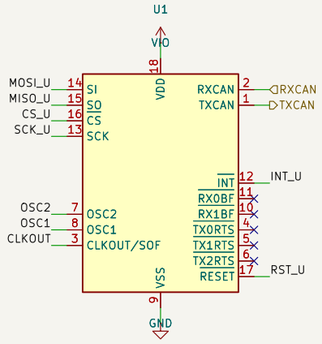
J2: 1 VIO, 2 GND, 3 SCK, 4 MOSI, 5 MISO, 6 CS, 7 INT, 8 RST





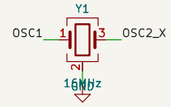
(3) MCP2515 CAN controller



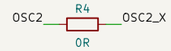


Pins 4–6 (TXnRTS) / 10–11 (RXnBF) unused: no-connect. CLKOUT to TP6 only (disable via CANCTRL.CLKEN=0).

(4) 16 MHz crystal oscillator









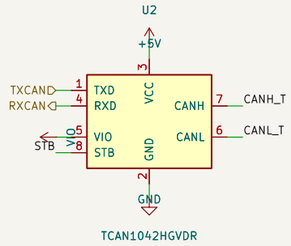
Sheet: 3 MCP2515 controller, 4 Crystal oscillator, 7 SPI host interface
9 V in · host-referenced VIO 3.0-5.5 V · JLCPCB

Sheet: /Controller/
File: controller.kicad_sch

Title: MCP2515 CAN Bus Module

Size: A3	Date: 2026-11-02	**Rev: A**
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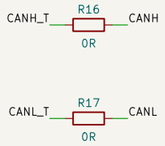
(5) CAN transceiver





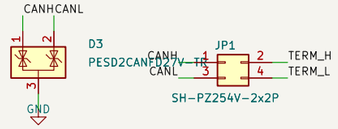
STB pulled low: normal mode always on.

(6) Bus protection, termination & connectors



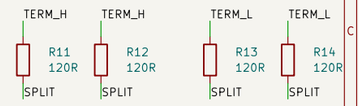


L2 DNP: if fitted, remove R16/R17.

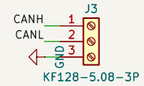


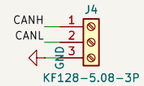
JP1: fit BOTH shunts (1–2, 3–4) = 120R split termination ON.











Sheet: 5 CAN transceiver, 6 Bus protection, termination & connectors
9 V in · host-referenced VIO 3.0-5.5 V · JLCPCB

Sheet: /CAN PHY/
File: can_phy.kicad_sch

Title: MCP2515 CAN Bus Module

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 4/4